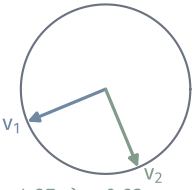
第一步：形成 Gram 矩阵

第二步：正交对角化，得特征方向

第三步：单位圆 → 椭圆

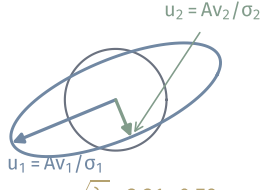
椭圆：$x^\top A^\top A x = 2$

$$A = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix} \qquad A^\top A = \begin{bmatrix} 4.25 & 1.5 \\ 1.5 & 1.25 \end{bmatrix}$$



$\lambda_1 \approx 4.87,\ \lambda_2 \approx 0.63$

$A^\top A = V\,\mathrm{diag}(\lambda_1, \lambda_2)V^\top$



$\sigma_i = \sqrt{\lambda_i} \approx 2.21,\ 0.79$

$A = U\,\mathrm{diag}(\sigma_i)V^\top$